Nested Lists 🪆

Christopher Martin - *Bowdoin College*
c.martin@bowdoin.edu

When we first talked about lists and tuples, we saw how we can store anything inside of them!

Today, we are going to see how 'anything' even means other lists / tuples!

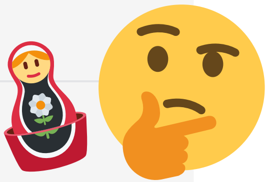
We can store lists and tuples within other lists and tuples!

When we do this, we say we are created 'nested' or multi-dimensional data!

```python
# To make them easier to understand, we try to space out the definition
grid = [
    ["A", "B", "C"],
    ["D", "E", "F"],
    ["G", "H", "I"]
]

print(grid)         # We can use the entire structure
print(grid[1])      # Just one of the nested lists
print(grid[1][2])   # Or one element within one of the nested lists!
```

```
[['A', 'B', 'C'], ['D', 'E', 'F'], ['G', 'H', 'I']]
['D', 'E', 'F']
```



Understanding a 2D List

Sometimes, we will call the special case of a 2D list / tuple a matrix!

We always access into a matrix first by row and then by column!

```python
grid = [
# Column:
#   0     1     2     Row:
  ["A", "B", "C"], # 0
  ["D", "E", "F"], # 1
]

print(grid)       # Access the entire grid
print(grid[1])    # Access row 1 of the grid
print(grid[1][2]) # Access column 2 of row 1 of the grid
```

```
[['A', 'B', 'C'], ['D', 'E', 'F']]
['D', 'E', 'F']
F
```

Building up a nested list

Adding a list / tuple to a list will cause a nested list to be formed!

```python
my_list = ["a", "b", "c"]
my_number_list = [1, 2, 3]

my_list.append(my_number_list)

print(my_list)

print(len(my_list)) # len only counts the surface-level elements!
print(2 in my_list) # in only checks the surface-level elements!
```

```
['a', 'b', 'c', [1, 2, 3]]
4
False
```

3D... 4D! Even 5D!

There is no limitation to how deeply you can nest lists!

However, we typically don't go above 2D or 3D very often...

```
grid = [
# Column:
#      0          1              2    Row:
  [[1, 2, 3], [4, 5, 6], [7, 8, 9, 0]], # 0
  [["A", "B", "C"], ["D", "E"], ["F"]], # 1
  [4.5, [3.2, 7.1], [[4.3, 5.6], 3.1]], # 2
]
```

```
No output received
```